



Determination of Total Carbon in Cast Iron / Steel by Strohlein Apparatus (Combustion Method) MME408



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Introduction:

A quick method, Strohlein apparatus, can be used to directly measure the carbon percentage in steel. Total carbon is equivalent to C in graphite and other sites. Here, basically, C is oxidized at high temperature in the presence of pure O_2 and yields CO_2 . Here, lots of other gases can be produced, like CO_2 , SO_2 , and excess O_2 . But CO_2 can be dissolved in KOH and forms a saturated solution. After that, volume change can be measured from the decrease in CO_2 in the gas content. Hence, the C percentage can be figured out from it, also taking the help of a correction factor which is mainly dependent on atmospheric pressure and temperature.

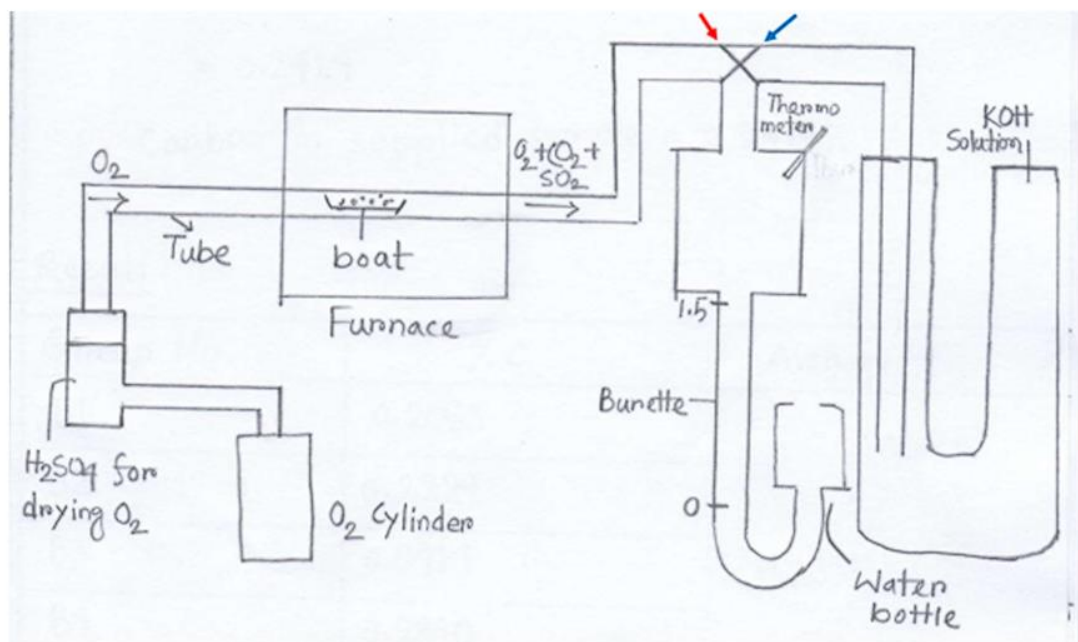


Figure 1: Strohlein Apparatus

Apparatus:

- 1g steel chips
- Ceramic boat
- O_2 Cylinder
- H_2SO_4
- KOH
- Strohlein Apparatus

Procedure:

- 1) First, 1g steel chip was collected from lab instructors.
- 2) Then it was transferred to a ceramic boat. Sn will be added inside the ceramic boat whose impact was discussed in the discussion section.
- 3) Then the boat was gone through the tube furnace.
- 4) Pure O_2 was supplied to the steel.

- 5) The gas was collected in the measuring burette(The solution inside tube was methyl red) and reading-1 was noted.
- 6) There was a cork in the absorption vessel which could be adjusted to let the gases meet the KOH in the other chamber.
- 7) Then after a certain time, the gas was again collected in the measuring burette chamber using the cork, and reading 2 was noted.



Figure 4: Ceramic boat

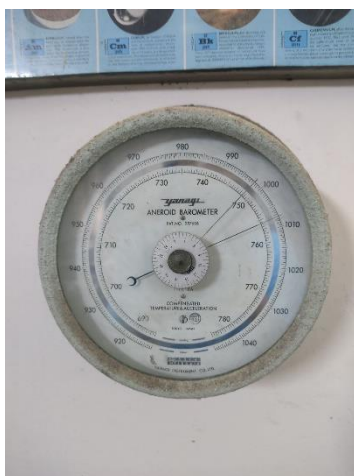


Figure 3: Barometer



Figure 2: O₂ Cylinder



Figure 5: Tube Furnace



Figure 6: Strohlein Apparatus

Result and Calculation:

Reading-1, $R_1 = 0.11$

Reading-2, $R_2 = 0.465$

Sample weight, $W = 1\text{ g}$

Temperature = 34°C ; Pressure = 756 mm(Hg)

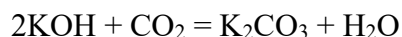
Correction factor = 0.903

Oxygen flow rate = $6\text{--}8\text{ m}^3/\text{s}$

$$\text{C percentage in steel} = \frac{R_2 - R_1}{\text{weight of sample}} \times \text{correction factor} = (0.465 - 0.11)/1 \times 0.903 = 0.321\%$$

Discussion:

- 1) Sn was used to decrease the melting temperature of steel. The melting temperature of steel is $1370 - 1540^\circ\text{C}$. Whereas when we used Sn, it **reduced the temperature** without impacting the reaction process.
- 2) H_2SO_4 was used to remove the moisture from O_2 , making it pure. Impure O_2 can damage the gas formation, hence overall calculation process.
- 3) Ceramic boat was used, because ceramic can withstand high temperature without any kind of altering its original properties. Moreover, it does not segregates at high temperature, so extra impurity treatment would not be a case to consider. In addition, **ceramic boat has large surface area which ensures easily reaction** and produce gaseous substances.
- 4) KOH will react with CO_2 and will form:



Here CO_2 will be dissolved and excess gas will go through the **measuring burette** and give a reading difference.

From the lab instructor, I learned that after a certain time, this **KOH will change its color to red**, indicating that the solution should be changed.

5) As our C percentage is 0.321%, it is a **medium carbon steel**. From the phase diagram :

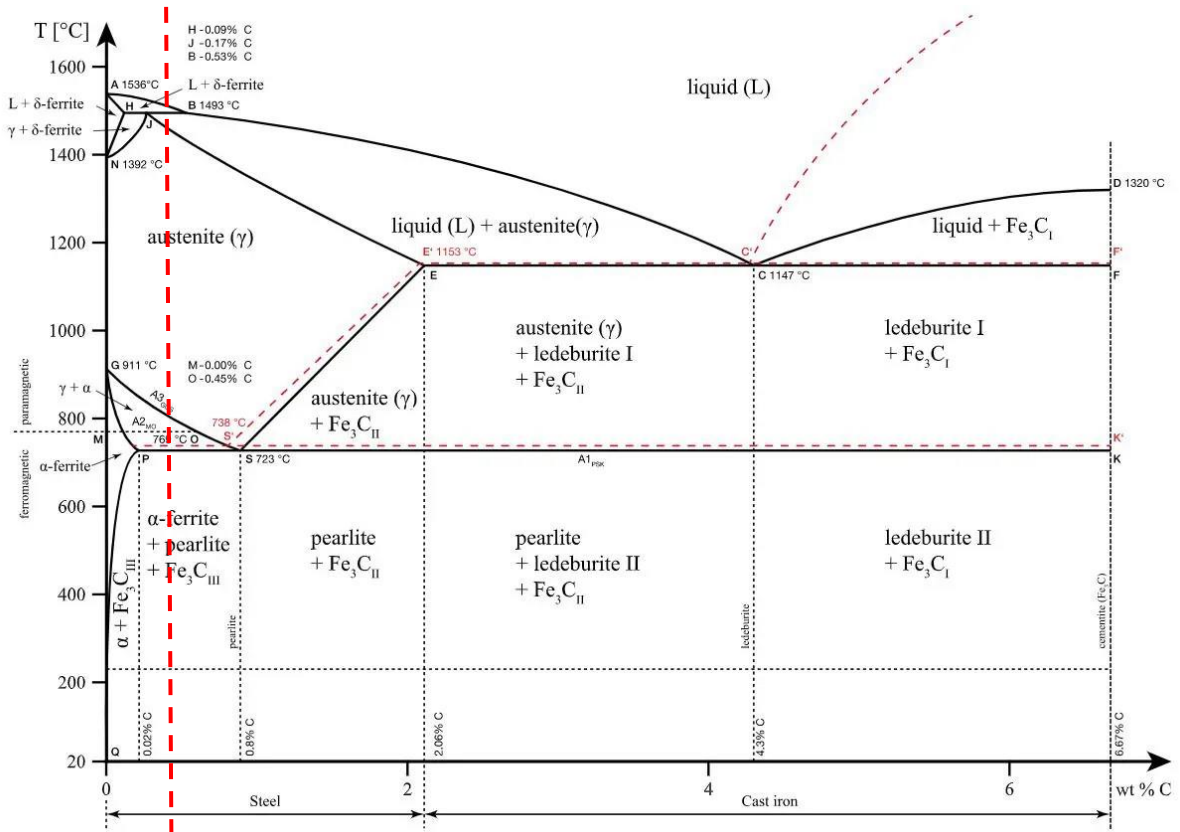


Figure 7: Fe-C Phase Diagram

It is in the region of hypo-eutectoid alloy. Microstructure of this region contains α -ferrite and cementite. As temperature decreases, austenite transforms to ferrite and alternate layers of ferrite and cementite. Below 723 °C the outside white layers are ferrite and inside in alternate layer the white portions are ferrite and grey portions are cementite.

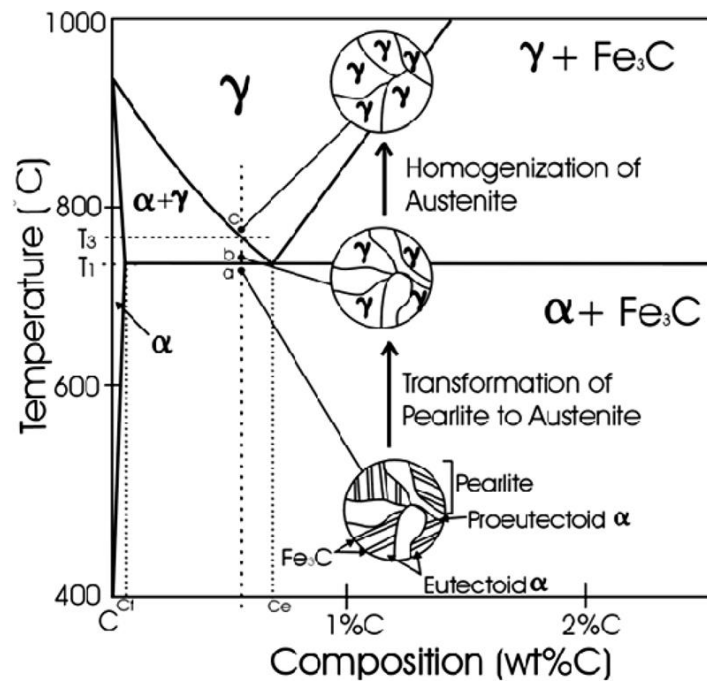


Figure 8: Microstructure of hypo-eutectoid alloy.

Medium carbon steel can be used in various applications like gears, crankshafts, rail wheels etc.

Data of other groups (steel) samples:

Group	Carbon (%)	Classification
Group 1	0.230	Low-carbon steel
Group 3	0.321	Medium-carbon steel
Group 4	0.298	Medium-carbon steel
Group 6	0.307	Medium-carbon steel

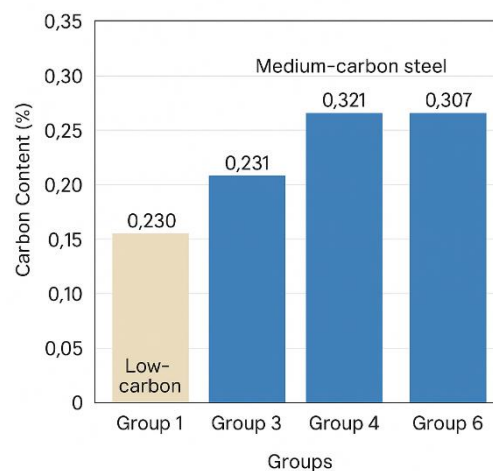


Figure 9: Data of different groups (steel)